

Urvashi Taki

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SUMMARY

Graduate AI Engineer specializing in computer vision, generative AI, and MLOps. Proven track record designing and deploying production-grade ML systems-from UAV surveillance pipelines to sensor-powered predictive maintenance-using Python, SQL, FastAPI, and cloud-native services. Skilled in CI/CD, model versioning, and real-time inference, delivering AI solutions that drive operational efficiency.

EDUCATION

University of Washington | *Master of Science, Electrical and Computer Engineering with a focus in AI/ML* (GPA: 3.92) Seattle, WA Sep 2024 - Jun 2026
MKSSS's Cummins College of Engineering | *Bachelors of Technology, Instrumentation and Control* (GPA: 3.5) Pune, India Aug 2017 - May 2021

TECHNICAL SKILLS

- Generative AI:** LLMs, Agentic Workflows, RAG Pipelines, Prompt Engineering, Evaluation Frameworks(RAGAS, G-Eval), LangGraph, CrewAI, Claude Code, OpenAI API
- ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Computer Vision (OpenCV), Neural Networks, CUDA, Machine Learning, JAX, Natural Language Processing, Optimization
- Data & Analytics:** Pandas, NumPy, Matplotlib, SQL, Vector Databases (Pinecone, Milvus, ChromaDB), ETL
- MLOps & Cloud:** AWS, GCP, Azure, Docker, Kubernetes, CI/CD, MLflow, Weights & Biases, Observability
- Software & Backend:** Python, C++, FastAPI, Flask, REST APIs, Linux/Bash, Git

RELEVANT EXPERIENCE

University of Washington | *Graduate Teaching Assistant* Sep 2025 - Mar 2026

EEP 596 (Fall 2025); EEP 596 A: LLM: Transformers to GPT (Winter 2026)

- Led instruction for two graduate and an undergrad-level AI courses (85+ students including industry professionals) covering transformers, retrieval-augmented generation, LLM fine-tuning, and core natural language processing concepts; designed curriculum and coding assignments adopted across three course offerings
- Led bi-weekly technical review sessions guiding student research groups through critical analysis and hands-on implementation of state-of-the-art papers, including attention mechanism variants and RLHF pipelines
- Designed coding assignments on transformer architectures and RAG pipelines; resolved complex implementation bugs in student PyTorch and HuggingFace projects during weekly office hours

Actuate | *Applied Computer Vision Research Intern* Jun 2025 - Dec 2025

- Developed and benchmarked a machine learning object-tracking pipeline evaluated on HOTA/MOTA metrics, achieving an 85% reduction in redundant ID assignments over the baseline and preventing \$10K in insurance losses per incident
- Extended GitHub Actions CI/CD workflows to automate model validation and deployment gates, integrating MLflow for experiment tracking and model versioning, reducing manual release overhead and enabling reliable rollouts across 100K+ CCTV stream deployments
- Extended the tracking framework to build a line-crossing detection system, translating product requirements into a FastAPI microservice for real-time alerting logic deployed across large-scale concurrent streams

Aerogravity Pvt Ltd (NxtQube) | *AI Engineer* May 2023 - Jul 2024

- Led a 6-engineer team to design and deploy a UAV surveillance system with real-time object detection and tracking; delivered production-ready capabilities on schedule
- Designed inventory tracking algorithm combining OCR and detection methods, achieving 96% accuracy - a significant improvement over the manual-inspection baseline
- Reduced operational overhead by 70% by migrating end-to-end AI stacks to AWS Lambda and Fargate
- Built real-time ETL pipelines processing 1M+ sensor datapoints/day, powering predictive maintenance models and reducing unplanned downtime

Aivara Technologies Pvt Ltd | *Founding AI Engineer, Co-Founder* Apr 2022 - Apr 2023

- Architected and patented (No. 202221025746) a real-time AI water quality monitoring system using diatom-based biosensors, edge ML inference, and IoT hardware integration - built from scratch with minimal funding
- Independently researched diatom taxonomy and biosensor literature to design the biological sensing layer, bridging microbiology domain knowledge with ML classification for a novel cross-disciplinary system
- Defined technology roadmap and scoped MVP under tight resource constraints; chose edge inference over cloud to eliminate connectivity dependency in remote deployment environments

PROJECTS

Foreseer - Beauty Brand Trend Intelligence Pipeline Mar 2026

- Designed an end-to-end GPT-4o extraction pipeline in Python, deployed as a FastAPI microservice on Azure using Docker, with MLflow experiment tracking and multi-layer schema enforcement to catch malformed field types and out-of-range values
- Built a keyword ontology engine using MiniLM-L6-v2 embeddings and agglomerative clustering to generate 640 weighted brand-specific keywords across six content universe, scored on relevance, brand alignment, trend velocity, and search volume
- Built a multi-source trend intelligence system integrating SerpApi (Google News + Shopping) with embedding-based relevance filtering, sponsored-content detection, and structured LLM extraction, producing a normalized 10-field trend object

Agentic Financial Q&A - Multi-Architecture Benchmark & Deployment Feb 2026

- Architected a multi-agent financial Q&A system with an orchestrator-critic pipeline, integrated 7 live tools (AlphaVantage, yfinance, SQLite) and tracked experiments and model versions using Weights & Biases to isolate silent hallucination failure modes
- Benchmarked baseline, single-agent, and multi-agent LLM architectures on 15 real-time financial tasks, quantifying a 73% accuracy gain from tool-calling and a measurable critic-layer tradeoff: zero hallucinations at the cost of 13 accuracy points
- Implemented stateful conversational memory across multi-turn financial queries with dynamic model routing between gpt-4o-mini and gpt-4o based on query complexity, and deployed as a Dockerized FastAPI service on AWS with CI/CD pipelines